Beyond Borders: Evaluating the Efficacy of Sub-National Informational Comparisons

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Electoral accountability depends on information

- Accountability: voters use elections to reward or punish incumbents based on performance

- For accountability to work, voters must be able to:
  - Observe what the government has done
  - Attribute outcomes to incumbent actions
  - Act on that assessment at the ballot box

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- Attribution is difficult when government performance is noisy

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- Many potential reasons for mixed findings
- Relatively less attention on how the information is presented

Does the way the information is presented matter?

- Studies typically vary *whether* information is provided, not *how* it is framed
- Real-world campaigns routinely use comparisons (Grossman and Slough 2022)
- How do comparisons change interpretation of information?

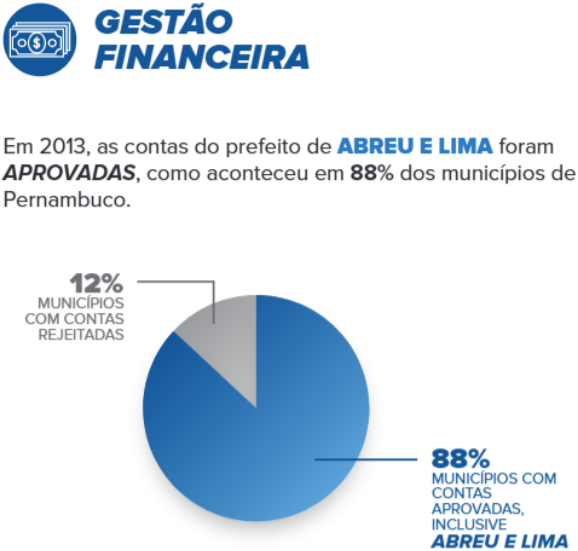


Figure 1: Informational leaflet from Boas, Hidalgo, and Melo (2019)

How might voters use comparative information?

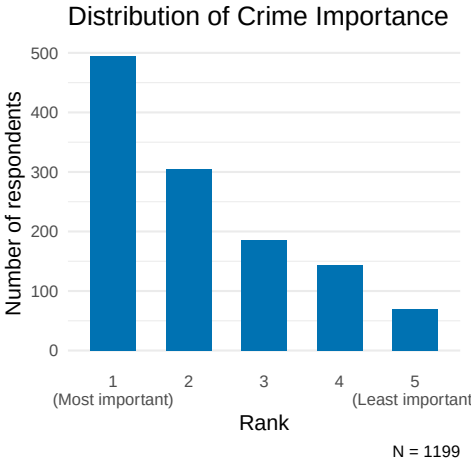
1. **Filtering noise:** comparisons clarify incumbent signals (Besley and Case 1995)
   - Relative to similar politicians and/or those with similar extenuating circumstances

2. **Constructing counterfactuals:** incumbent to opposition comparisons

3. **Unhelpful complexity:** comparisons confuse more than help (Arias et al. 2024)

Survey in Mexico to examine effect of comparisons

- Two-wave online survey, $n \approx 2{,}180$
- Performance information: real robbery data for respondent's own municipality
- Crime is a top voter concern; municipal governments control police
- Two components: experimental and descriptive

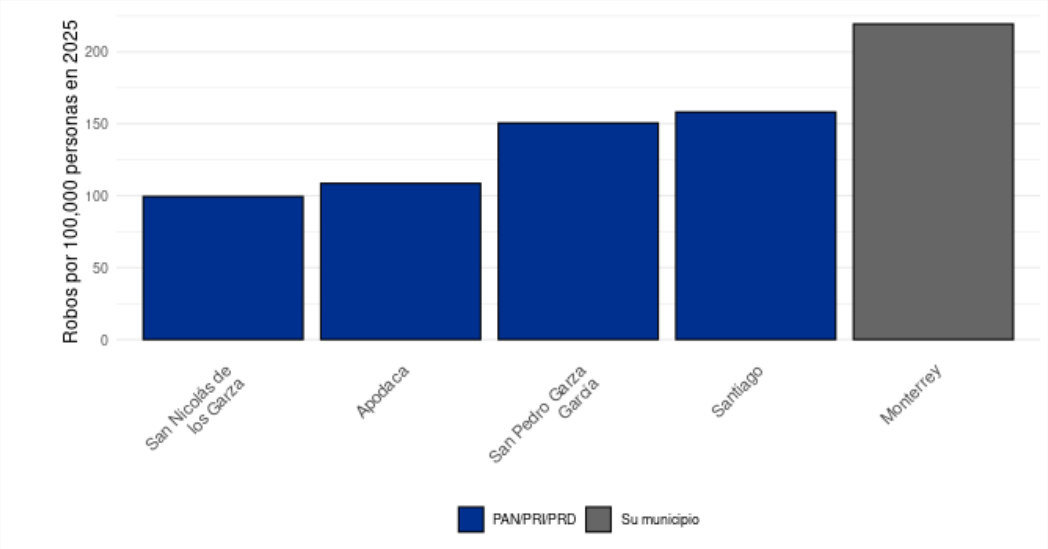


Experimental Design

Four treatment arms vary comparisons are shown

Arm	What the respondent sees
Plain info	Municipality robbery statistics
Non-partisan	Plain info + comparison to similar municipalities
Out-coalition	Plain info + comparison to opposition-led municipalities
Same-coalition	Plain info + comparison to same-coalition municipalities

- Comparisons matched on population, area, and distance
- Control groups receive data on weather patterns instead of crime
- Outcomes: incumbent crime evaluation and prospective vote choice



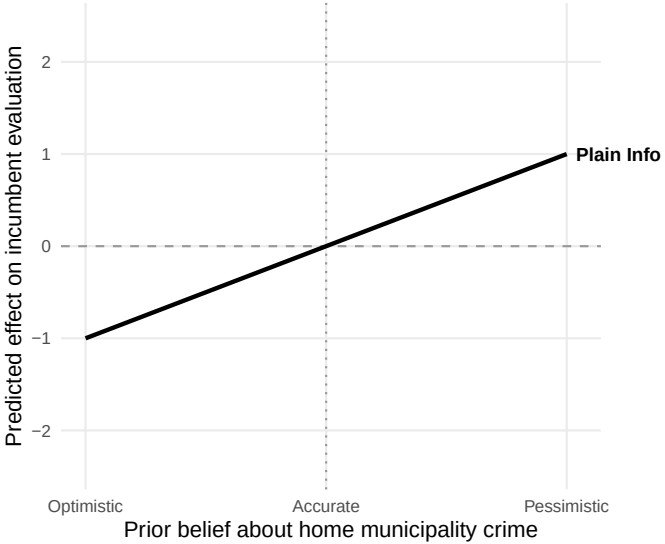
Fuente: Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública (SESNSP).

Focus on heterogeneous treatment effects

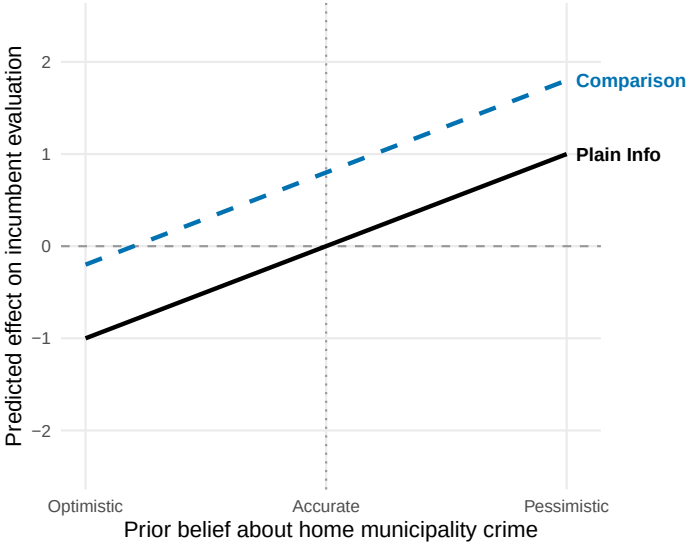
- Treatment effects should differ based on content of info and respondent priors
- Voters who are pessimistic will receive “good news” and vice versa
- For comparison arms, relative performance may further shift evaluations

Equation

Plain information updating



Favorable peer comparison



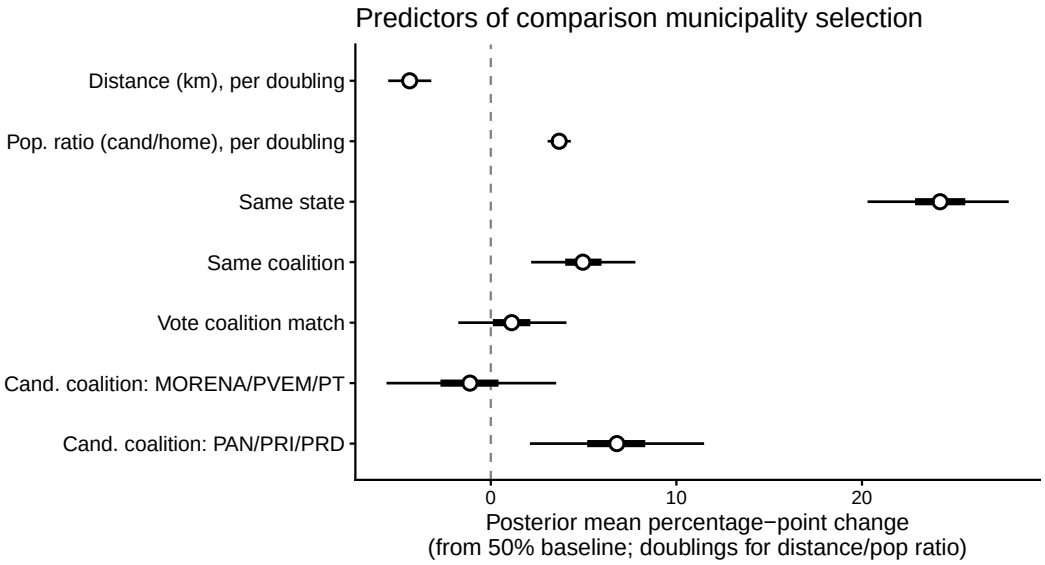
Descriptive Analysis

How do voters choose comparison municipalities?

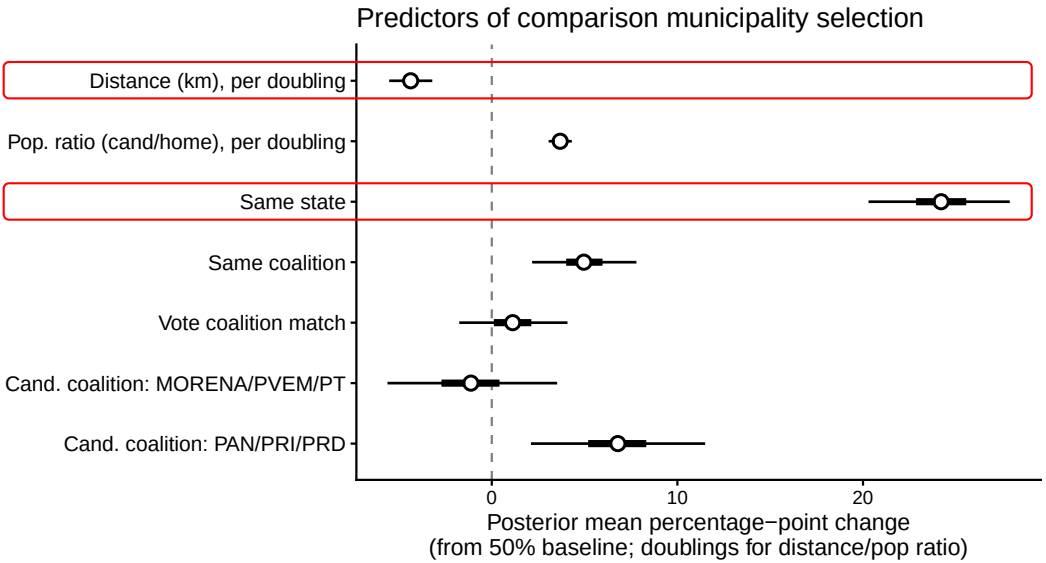
- Shown 15 candidate municipalities before treatment
- Select any they would use as a comparison to their own
- Candidates drawn from three pools: nearby, nationally largest, random
- Bayesian multilevel logistic regression to analyze selection patterns Equation

Governments are often judged by comparing them to others. If you could choose which municipalities to compare to Monterrey, Nuevo León, which would you pick?

- ☐ Acuña, Coahuila de Zaragoza
- ☐ Atemajac de Brizuela, Jalisco
- ☐ Chimalhuacán, México
- ☐ Naucalpan de Juárez, México
- ☐ Abasolo, Nuevo León
- ☐ General Escobedo, Nuevo León
- ☐ Guadalupe, Nuevo León
- ☐ El Carmen, Nuevo León
- ☐ San Nicolás de los Garza, Nuevo León
- ☐ San Marcial Ozolotepec, Oaxaca
- ☐ Santiago Laxopa, Oaxaca
- ☐ Puebla, Puebla
- ☐ San Luis Potosí, San Luis Potosí
- ☐ Hermosillo, Sonora
- ☐ Abasolo, Tamaulipas



N = 1199 respondents. Thick lines: 50% CI. Thin lines: 95% CI.



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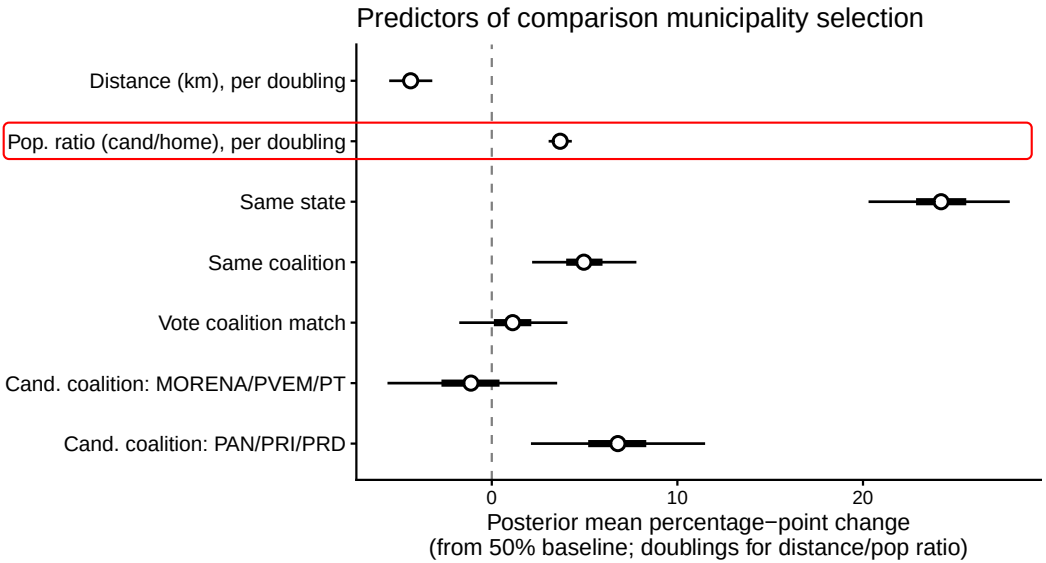

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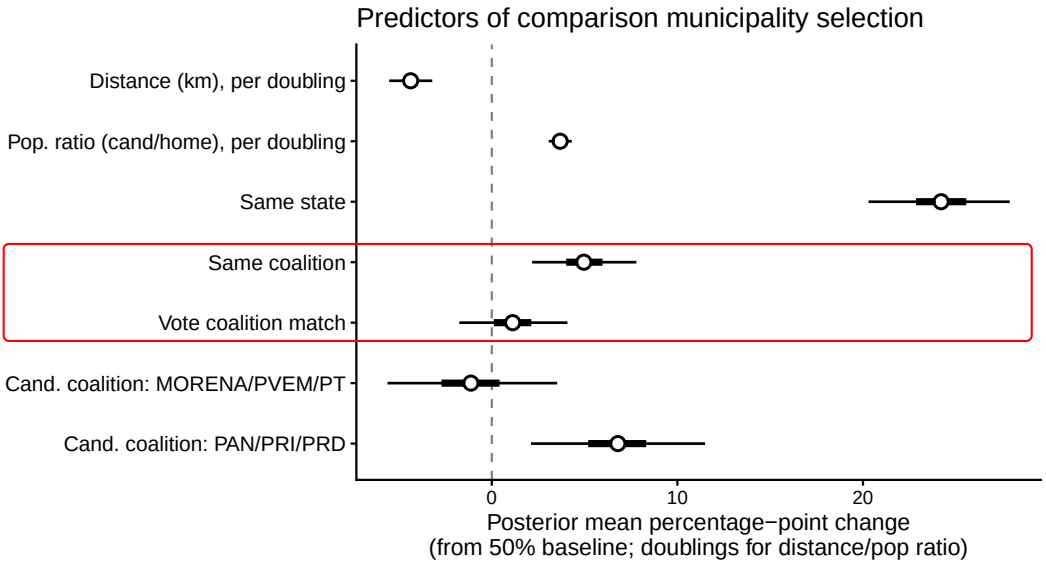

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Thank You

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Treatment Effect Estimating Equation

$$y_i = \alpha_0 + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=0}^{4} \delta_k T_{ik} + \underbrace{\sum_{k=0}^{4} \beta_k T_{ik} \times CG_i}_{\text{Treatment effect conditional on content of info}} + \underbrace{\sum_{k=0}^{4} \gamma_k T_{ik} \times RG_i}_{\text{Treatment effect conditional on peer ranking}} + \varepsilon_i$$

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Bayesian multilevel logistic regression

$$\Pr(Y_{ri} = 1) = \text{logit}^{-1}(\mathbf{X}_{ri}\boldsymbol{\beta} + \alpha_r)$$

- $Y_{ri} = 1$ if respondent r selects municipality i
- $\mathbf{X}_{ri}$: municipality-level covariates
- α_r: respondent-level random intercept

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